Learning Association-Line Transformations over Autosegmental Graphs

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Rutgers Subregular Workshop · November 7, 2026

Background

- Computational models of tone have largely relied on string-based representations.
- Autosegmental representations (ARs) explicitly encode tones, tone-bearing units (TBUs), and their associations.
- Work on tonotactic learning (Li & Heinz) provides a foundation for learning over ARs.
- Work on logical transductions (Yolyan 2025) provides a foundation for learning processes over model-theoretic representations.

Today

- A method for learning tonal processes over ARs, illustrated with contour formation triggered by vowel deletion.
- It learns the logical conditions for **linking** and **delinking** in length-preserving AR mappings.
- Each condition is found by separating environments where a change happens (positive evidence) from environments where it does not (negative evidence).

Tonal Processes Triggered by Vowel Deletion

A mini typology: Engenni, Etsako, Ogoja Yala

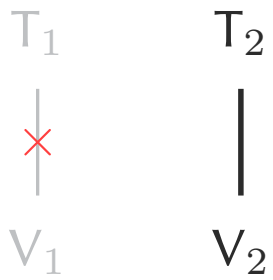
Tone deletion in Engenni

In vowel hiatus, Engenni deletes V_1 and also removes its tone.

L H H

/ɔ̀kò # édèì/ → [ɔ̀kédèì]

man canoe → ‘a man’s canoe’



H L L

/fùnú # èdhí/ → [fùnèdhí]

climb palm.tree → ‘to climb a palm tree’

Contour formation in Etsako

In vowel hiatus, Etsako deletes V_1; its tone reassociates to the surviving vowel to form a contour.

L H LH

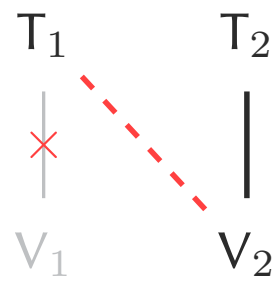
/ówà # ówà/ → [ówǒwà]

house house → ‘every house’

H L HL

/dé # àkpà/ → [dâkpà]

buy cup → ‘buy a cup’



V_2 now bears T_1 and T_2

Higher-pitch preservation in Ogoja Yala

Ogoja Yala deletes V_1; the higher tone (H, else M) survives on V_2.

L H H
/dè # áchí/ → [dáchí]

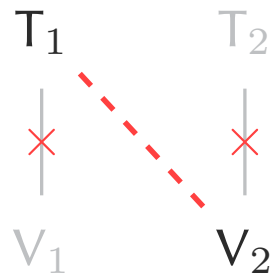
give grass → ‘to give grass’

H L H
/má # ɔ̀chí/ → [mɔ́chí]

see tree → ‘to see a tree’

M L M
/bī # ɔ̀chí/ → [bɔ̄chí]

carry stick → ‘to carry a stick’



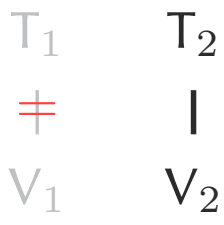
When T_1 is higher: two lines removed, one added. Otherwise as in Engenni.

Two operations: delink and link

All three languages delete V_1; they differ in which association lines change.

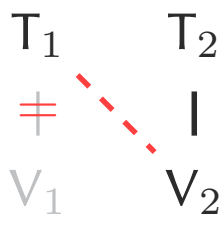
Engenni

delink



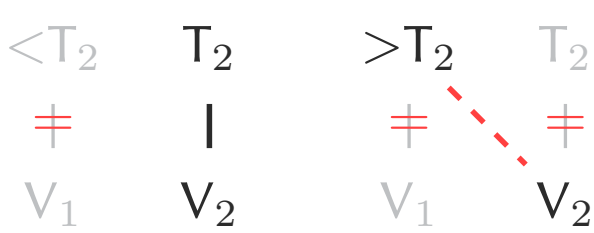
Etsako

delink + link



Ogoja Yala

delink + link, if T_1 is higher



Every outcome is built from two operations on association lines: **delink** (=) and **link** (dashed).

The Proposal

One transduction, two learnable predicates

Logical Transduction

Output		Input	Interpretation
ϕ_{Domain}	$:=$	True	Preserve all input elements
Copy	$:=$	1	Define output domain same size as input
$\phi_P(x)$	$:=$	$P(x)$	Preserve elements and their labels
$\phi_<(x,y)$	$:=$	$<(x,y)$	Preserve linear order within tiers
$\phi_\alpha(x,y)$	$:=$	?	
$\phi_{\text{license}}(x)$	$:=$	?	

Table: Logical formulas defining an input-output AR transduction.

One transduction for all three languages

Tone x and TBU y are associated in the output iff

$$\phi_\alpha(x,y) := \underbrace{\big(\alpha(x,y) \wedge \neg \mathsf{delink}(x,y)\big)}_{\text{Keep}} \vee \underbrace{\big(\neg\alpha(x,y) \wedge \mathsf{link}(x,y)\big)}_{\text{Add}}$$

- **Keep** an input line unless delink holds.
- **Add** a new line when link holds.
- Everything else is copied: $\phi_P(x) := P(x)$ for each label P, and $\phi_\lhd(x,y) := x \lhd y$.
- A vowel left with no tone is not licensed and disappears:

$$\phi_{\mathsf{license}}(x) := C(y) \vee \exists x\, \phi_\alpha(x,y)$$

Learning task: learn delink and link from input–output pairs.

Representations and Evidence

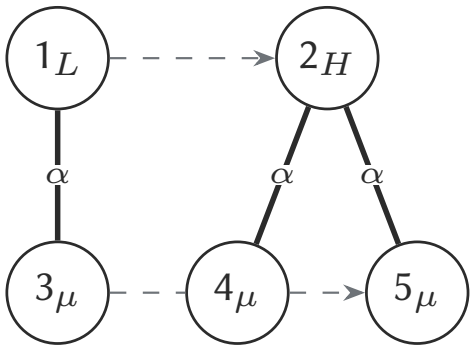
ARs as models, and what counts as a change

Autosegmental representations as models

AR



Graph



α = association $\lhd$ = successor

Model

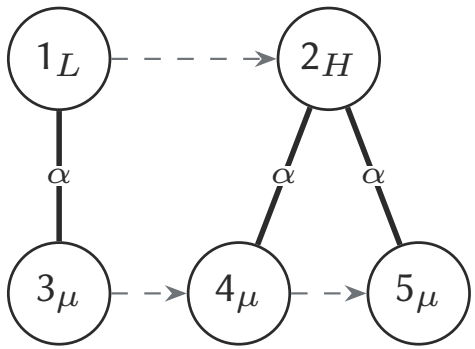
$$D = \{1, 2, 3, 4, 5\}$$
$$L = \{1\}$$
$$H = \{2\}$$
$$\mu = \{3, 4, 5\}$$
$$< = \{\langle 1, 2\rangle, \langle 3, 4\rangle, \langle 4, 5\rangle, \langle 3, 5\rangle\}$$
$$\alpha = \{\langle 1, 3\rangle, \langle 2, 4\rangle, \langle 2, 5\rangle\}$$

Three views of the same disyllabic LH.H word

Associations as a matrix

Rows are TBUs, columns are tones; a cell is 1 iff the two are associated.

Graph

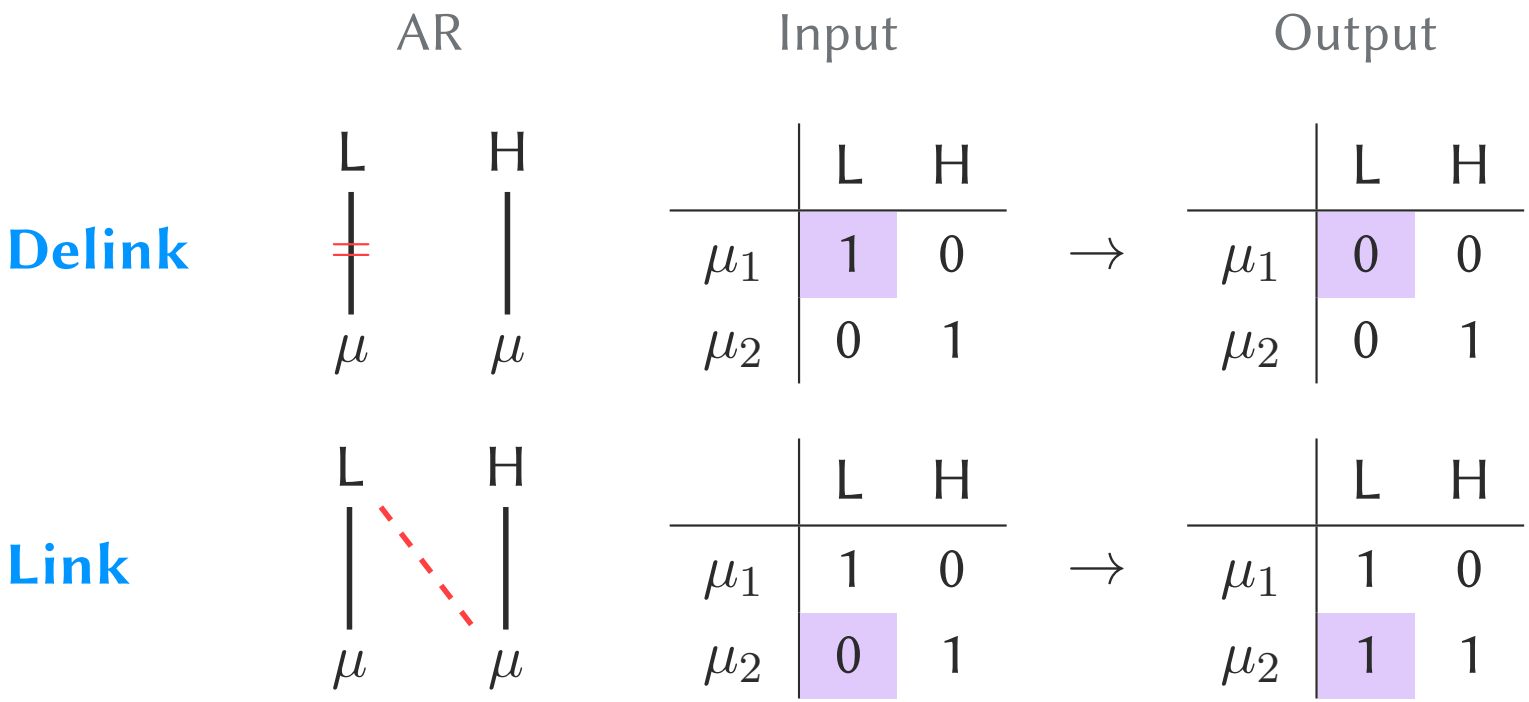


$\Longleftrightarrow$

Matrix

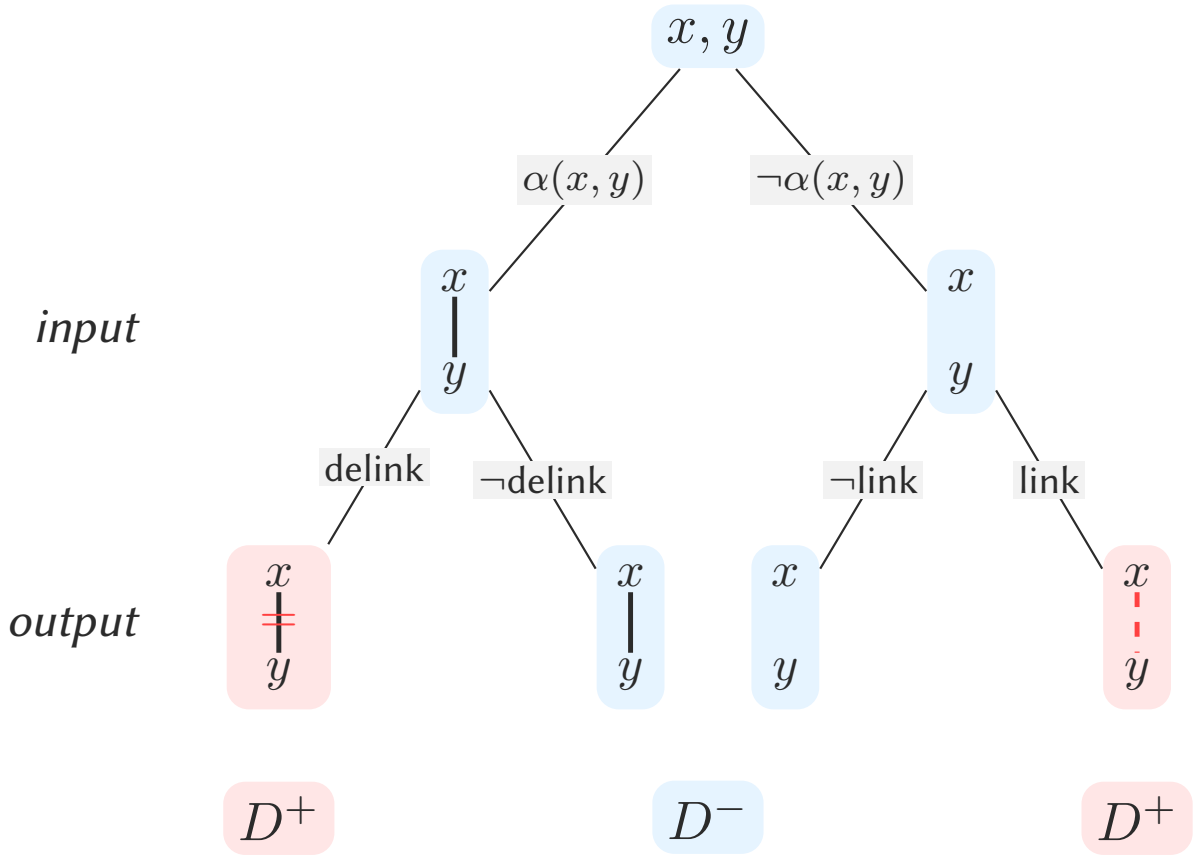
	L	H
μ_1	1	0
μ_2	0	1
μ_3	0	1

Association changes as matrix cells



Delinking: a cell goes $1 \rightarrow 0$. Linking: a cell goes $0 \rightarrow 1$.

Positive and negative evidence



D^+ a cell that changed: $\{ (I_i, x, y) \mid (I_i)_{xy} \neq (O_i)_{xy} \}$

D^- a cell that stayed the same: $\{ (I_i, x, y) \mid (I_i)_{xy} = (O_i)_{xy} \}$

The Learner

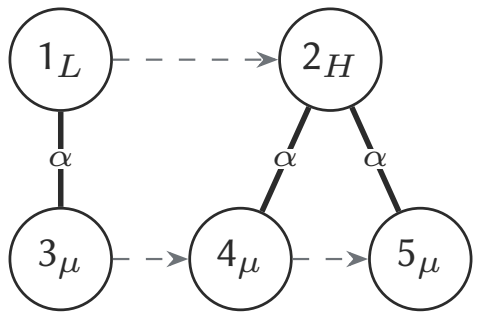
Searching a space of factors

Hypotheses are factors

A hypothesis is a small AR (a *factor*).

The *size* of an AR counts its nodes, labels, and association lines:

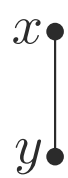
$$\text{size}(A) = |D| + \textstyle\sum_P |P| + |\alpha|$$



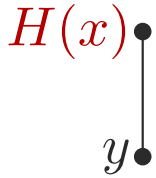
$$\text{size}(A) = \underset{|D|}{5} + \underset{|L|}{1} + \underset{|H|}{1} + \underset{|\mu|}{3} + \underset{|\alpha|}{3} = \mathbf{13}$$

The Next Superfactors: one step more specific

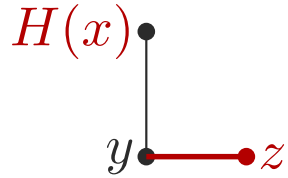
A *next superfactor* adds one piece to a factor, so it holds in fewer places.



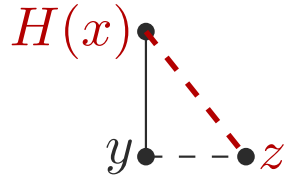
Start
size 3



Add label
size 4



Add node
size 5



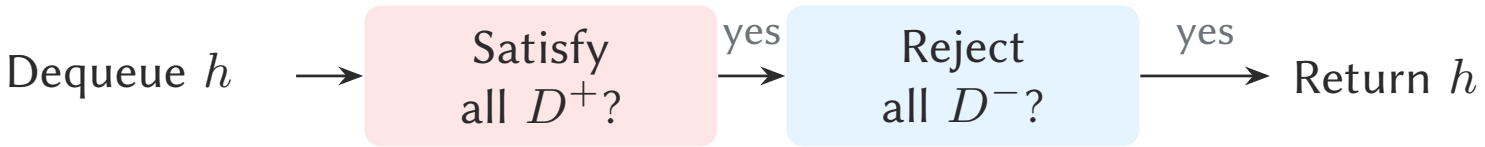
Add link
size 6

Convention: x is the tone (top), y the seg (bottom)

The learner

Start from the most general hypothesis for each predicate:

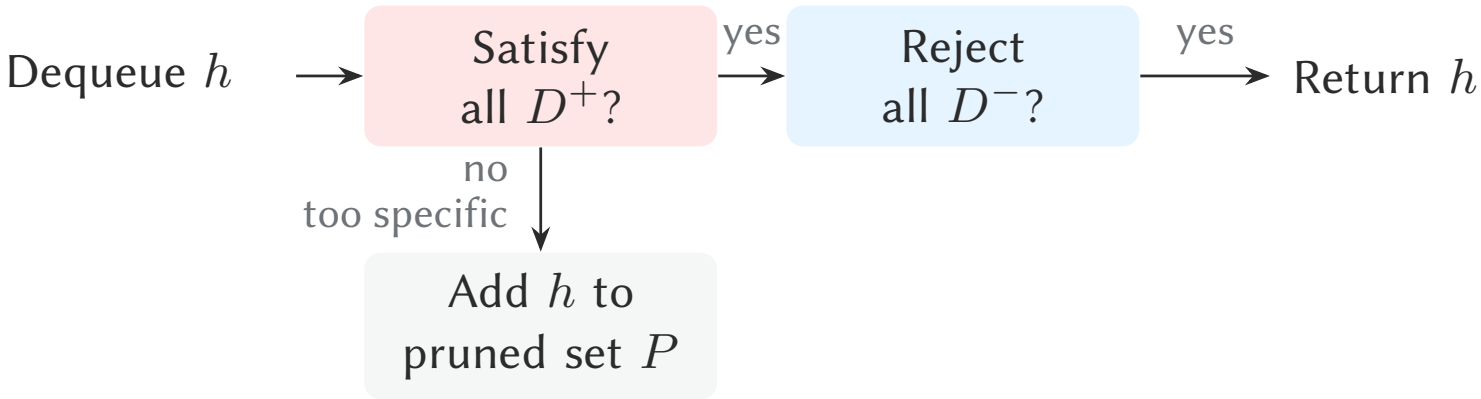
delink	$h_{d0} = \alpha(x, y)$	D^+: cells $1 \to 0$
link	$h_{\ell 0} = \neg\alpha(x, y)$	D^+: cells $0 \to 1$



The learner

Start from the most general hypothesis for each predicate:

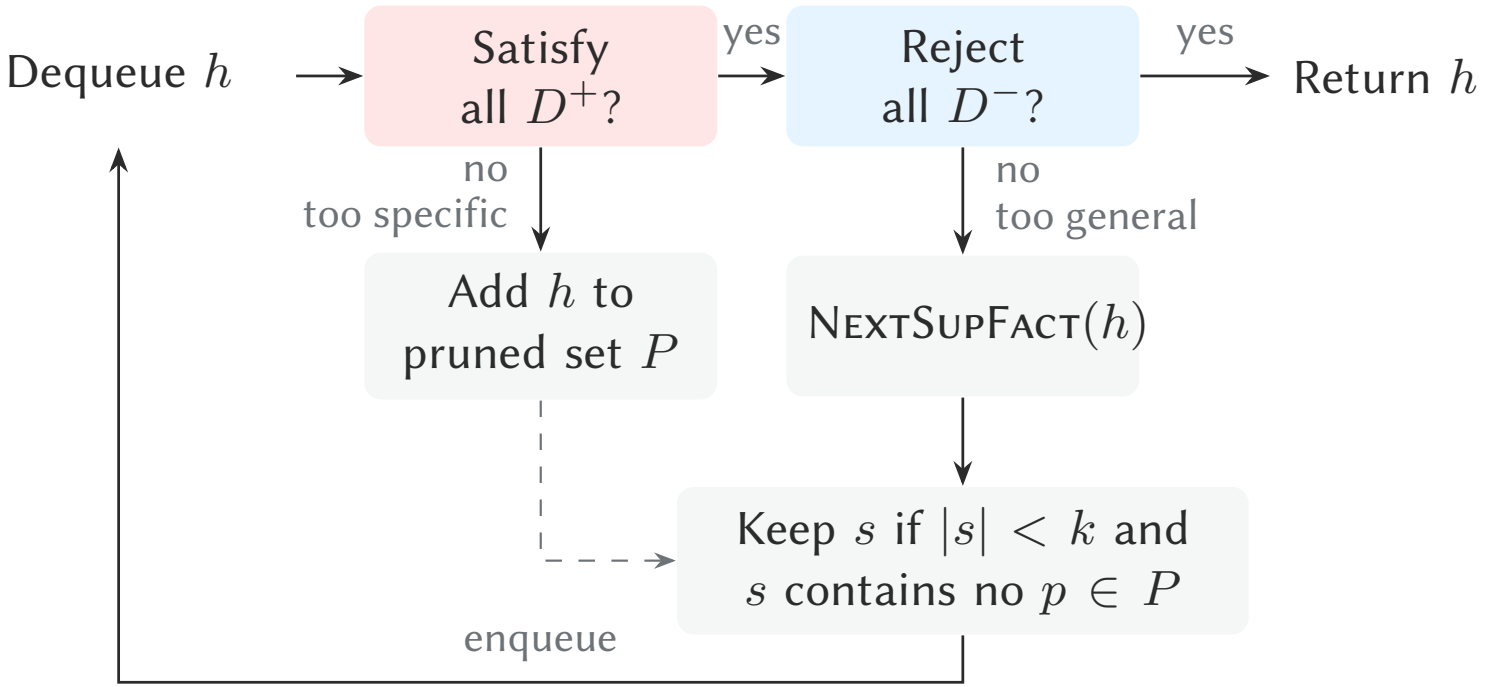
delink	$h_{d0} = \alpha(x, y)$	D^+: cells $1 \to 0$
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The learner

Start from the most general hypothesis for each predicate:

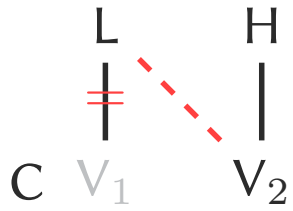
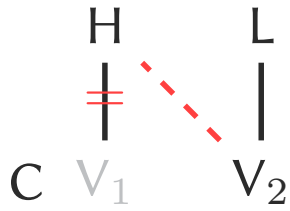
delink	$h_{d0} = \alpha(x, y)$	D^+: cells $1 \to 0$
link	$h_{\ell 0} = \neg\alpha(x, y)$	D^+: cells $0 \to 1$



Worked Example: Toy Data based on Etsako

Learning delink, then link

Toy data based on Etsako

	CV-H	CV-L	VCV-LH	CVV-LH	CVV-HL
UR-SR					

delink

- D^+ : two lines crossed out
- D^- : the six lines kept

link

- D^+ : two lines added (dashed)
- D^- : tone-segments unlinked

We view these as two independent processes and learn them separately.

Delink, step 0: $h_{d0} = \alpha(x, y)$

h_{d0} : delink$(x, y) = \alpha(x, y)$ "delink every line"

	CV-H	CV-L	VCV-LH	CVV-LH	CVV-HL
UR-SR					
D^+				✓	✓
D^-	×	×	× ×	×	×

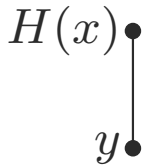
Covers both D^+ sites, but rejects none of the six D^- sites ⇒ **too general**.

✓ = good (covers a D^+ site / rejects a D^- site); × = bad.

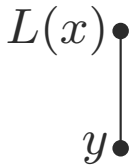
Delink, round 1: candidate superfactors

Superfactors of h_{d0} add one label or one node. Both nodes are already linked, so there is no site for a new association.

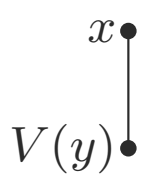
Add a label



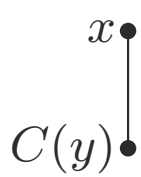
(a)



(b)

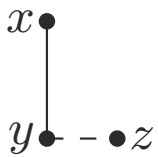


(c)

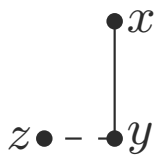


(d)

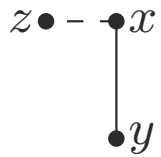
Add a node



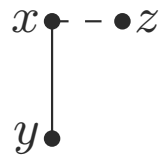
(e)



(f)

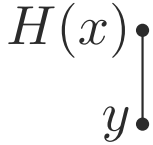
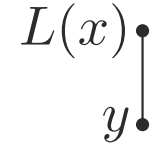
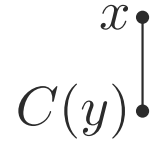
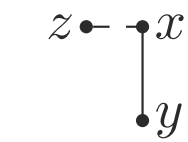


(g)

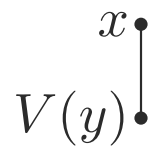
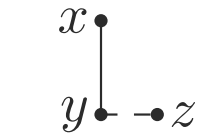
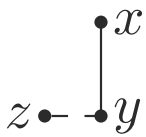
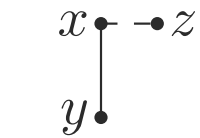


(h)

Round 1: too specific ⇒ pruned

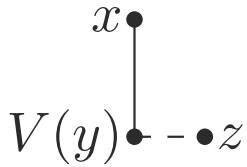
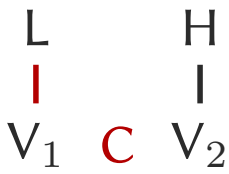
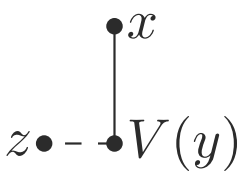
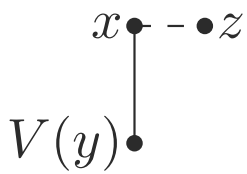
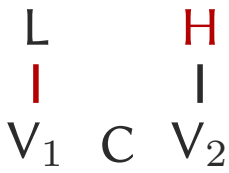
	Candidate	Why it fails D^+
(a)		misses CVV-LH: the delinked tone is L
(b)		misses CVV-HL: the delinked tone is H
(d)		misses both: consonants never bear tone
(g)		misses both: the delinked tone is first on its tier

Round 1: too general ⇒ NxtSupFac

	Candidate	D^- match	
(c)		H C V	every TBU is a V
(e)		L H V C V	V_1 is followed by C
(f)		H C V	V is preceded by C
(h)		L H V C V	L is followed by H

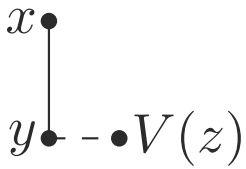
(c), (e), (f), (h) go back to NextSupFact; superfactors that contain (a), (b), (d) or (g) are discarded.

Delink, round 2: superfactors of (c)

	Candidate	D^+	D^- match
(c1)		✓	
(c2)		✓	
(c3)		✓	

Being a vowel is not what triggers delinking.

Delink, round 2: the consistent candidate

	Candidate	D^+	D^-	D^+ match
(e1)		✓	✓	L H | | C V_1 V_2

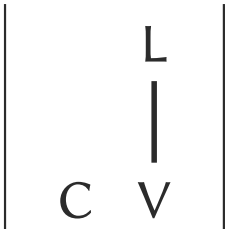
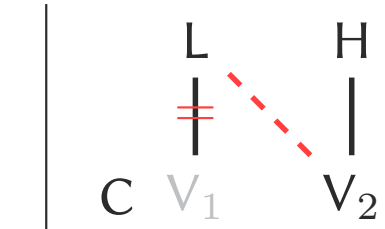
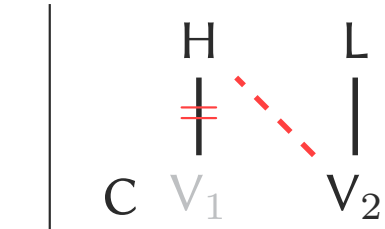
Covers both D^+ sites and rejects all six D^- sites ⇒ **return**:

$$\text{delink}(x, y) := \alpha(x, y) \wedge \exists z \left(y \lhd z \wedge V(z) \right)$$

Delinking is triggered by vowel hiatus: the TBU is followed by another vowel.

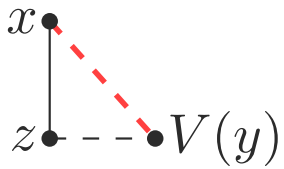
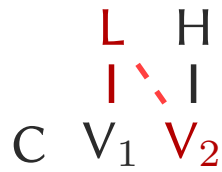
Link, step 0: $h_{\ell 0} = \neg\alpha(x, y)$

$h_{\ell 0}$: link$(x, y) = \neg\alpha(x, y)$ "link every unlinked pair"

	CV-H	CV-L	VCV-LH	CVV-LH	CVV-HL
UR-SR					
D^+				✓	✓
D^-			× ×	×	×

Covers both new lines, but rejects no unlinked pair ⇒ **too general**. The same search runs on its superfactors.

Link: the consistent candidate

Candidate	D^+	D^-	D^+ match
	✓	✓	

Covers both D^+ sites and rejects every D^- pair $\Rightarrow$ **return**:

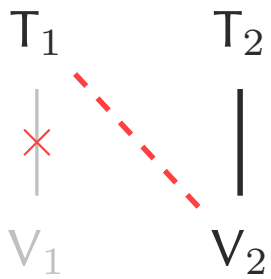
$$\text{link}(x, y) := \neg\alpha(x, y) \wedge V(y) \wedge \exists z \left(\alpha(x, z) \wedge z \lhd y\right)$$

A tone relinks to the vowel that follows its own TBU.

Putting it together: Etsako

Plugging both learned predicates into the transduction:

$$\phi_\alpha(x,y) := \underbrace{\alpha(x,y) \wedge \neg\exists z \left(y \lhd z \wedge V(z)\right)}_{\text{keep a line unless its vowel is in hiatus}}$$
$$\vee \underbrace{\neg\alpha(x,y) \wedge V(y) \wedge \exists z \left(\alpha(x,z) \wedge z \lhd y\right)}_{\text{add a line from a tone to the next vowel}}$$



V_1 loses its only tone, so ϕ_{license} deletes it; its tone is carried over to V_2, forming the contour.

Conclusion

- Tonal processes triggered by vowel deletion are changes to association lines: one transduction ϕ_α with two learnable predicates, delink and link.
- Learning a predicate = finding a factor that holds at every changed site (D^+) and at no unchanged site (D^-).
- Bottom-up search over superfactors: failing D^+ is to be pruned; failing D^- means NxtSupFac further.
- On the Etsako toy data the learner recovers both conditions:
  - delink: the TBU is followed by a vowel (vowel hiatus);
  - link: the tone's TBU is followed by this vowel.
- Every intermediate hypothesis is a small, readable AR.

Limitations

- **Ogoja Yala is not handled.** Which tone survives depends on the *relative* height of T_1 and T_2. T_2 is delinked (and T_1 relinked) only when T_1 is higher:

  $$(H \lhd M) \ \vee \ (H \lhd L) \ \vee \ (M \lhd L)$$

  This is a disjunction; no single factor covers all three D^+ cases without also matching D^- cases such as $L \lhd H$.

- Only length-preserving mappings; deletion is handled indirectly through licensing.

- Results so far are on toy data, and the learner is not yet implemented.